

Ex the space $P_1, B = [x, 1]$ | the space $\mathbb{R}^2 = [1, 0, e, 1]$

$$\begin{bmatrix} 2025 \\ 3\pi \end{bmatrix} = 2025x + 3\pi$$

$$\begin{bmatrix} 2025 \\ 3\pi \end{bmatrix} = (2025, 3\pi)$$

both are dim 2
and have scalars from $\mathbb{R}$

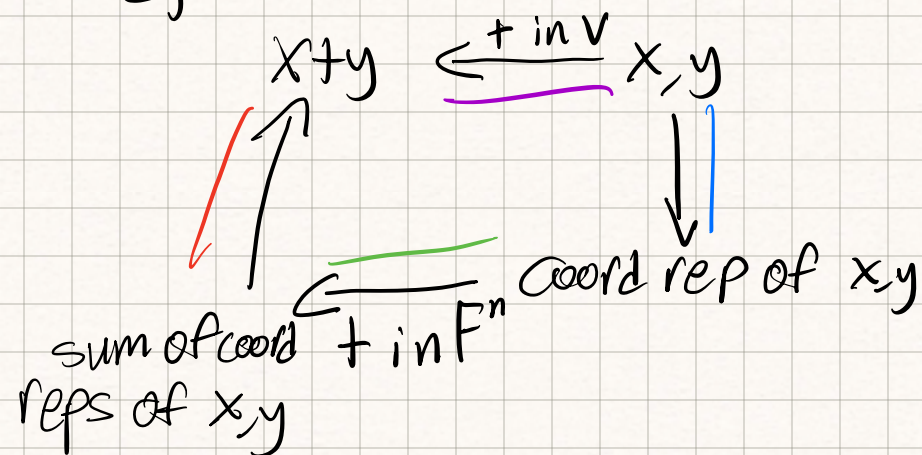
if V is finite dimensional $\dim V = |\text{basis}| = \# \text{ of coefficients in a basis rep.}$

Ex: $P_3 (\mathbb{R} \rightarrow \mathbb{R})$

$$\begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix}$$

Observations: if V is finite dimensional and we fix a basis of length n for V :

1) vector addition $\leftrightarrow$ addition in F^n



Ex $P_2 \mathbb{R} \rightarrow \mathbb{R} \quad B = [x^2, x, 1]$

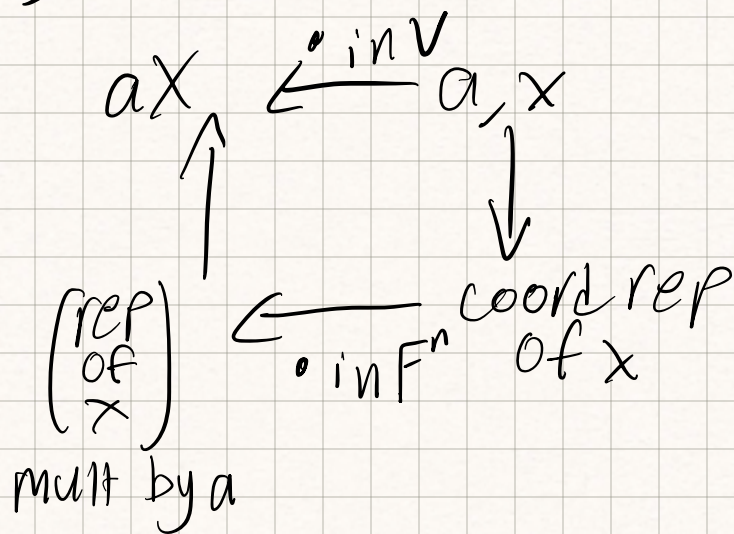
$$(x^2 - x) + (2x - 6) = x^2 + x - 6$$



$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ -6 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -6 \end{bmatrix} = x^2 + x - 6$$

$\mathbb{R}^3$

2) scalar mult $\longleftrightarrow$ scalar mult in F^n



Thm: let V be a vector space w/ finite dim n
let B be a basis for V

$$V \xrightarrow{R} F^n \quad R(v) = \begin{bmatrix} - \\ \vdots \\ - \end{bmatrix} \text{ is the unique rep of } v \text{ as a list of coefficients}$$

Then R is a linear function between vector spaces
e.g. $R(w+v) = R(w) + R(v)$ $R(av) = aR(v)$

R is surjective (onto)
and injective (1 to 1)

$x \rightarrow x^3$ is surjective
✗ and injective

$x \rightarrow x^2$ is not
surjective
✗ is not injective

Once we choose a basis for V , V essentially is the same as F^n
 $V \cong F^n$ EX: $P_3 \cong \mathbb{R}^4$

Thm: All vector spaces of the same dim are (in some sense) the same
w the same scalars

Ex: Consider the 3 vector spaces P_1 , $\mathbb{R}^2$, $\hookrightarrow$
if we fix a basis, these should all be the same

$$P_1 \quad B_1 = [x, 1] \quad 3x - \tilde{\kappa} \Rightarrow \begin{bmatrix} 3 \\ -\tilde{\kappa} \end{bmatrix}$$

$$\mathbb{R}^2 \quad B_2 = [(1,0), (0,1)] \quad (3, -\tilde{\kappa}) \Rightarrow \begin{bmatrix} 3 \\ -\tilde{\kappa} \end{bmatrix}$$

$$\hookrightarrow B_3 = [\rightarrow, \uparrow] \quad \underbrace{3(\rightarrow) + (-\tilde{\kappa})(\uparrow)} \Rightarrow \begin{bmatrix} 3 \\ -\tilde{\kappa} \end{bmatrix}$$

